

Groin pain

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Soccer was my first love, and I enjoyed playing on the left wing until a groin injury forced me out of the game in 2008.

Chad le Clos, South African swimming gold medallist

This chapter discusses both acute and longstanding groin injuries in athletes. It covers the anatomy where the complexity and degree of interconnections is highlighted. Epidemiology and risk factors have been updated using systematic reviews. The 2015 Doha agreement meeting on terminology and definitions in groin pain in athletes is included and the chapter follows the terminology agreed upon during this expert meeting. Clinical examination and assessment is covered including a practical guideline for systematic assessment of an injured athlete. There is a new section on imaging highlighting the lack of evidence in this area. Treatment with specific sections on adductor, iliopsoas, inguinal and pubic-related groin pain as well as acute groin injuries is covered with suggestions for rehabilitation.

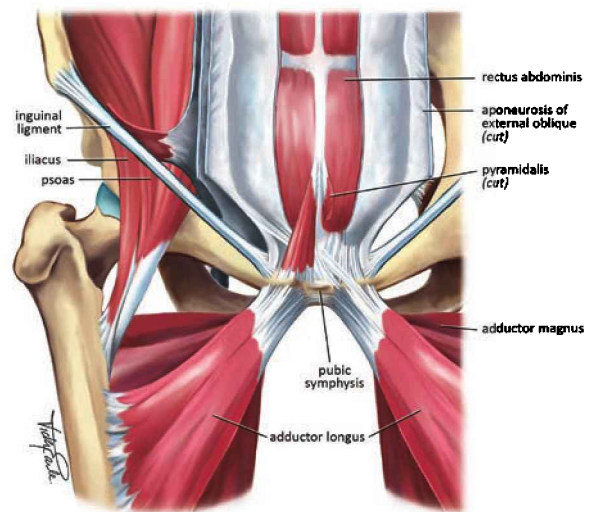
ANATOMY

The word groin most likely originates from the ancient English word *grynde* which meant abyss or void. Confusion surrounding the groin region has made clinicians refer to this region as the 'Bermuda triangle' of the body.¹ In this section, we explain and simplify the complex anatomy of the groin. The groin region, where the abdomen meets the lower limbs via the pelvis, contains not only musculoskeletal structures, but also important internal organs and urogenital structures. In this section we focus on the musculoskeletal anatomy of the groin. The anatomy of the hip joint is described in Chapter 31.

We will focus on the pubic symphysis, hip adductors, hip flexors, abdominals and the inguinal region. An overview of the anatomy is depicted in Figure 32.1a.

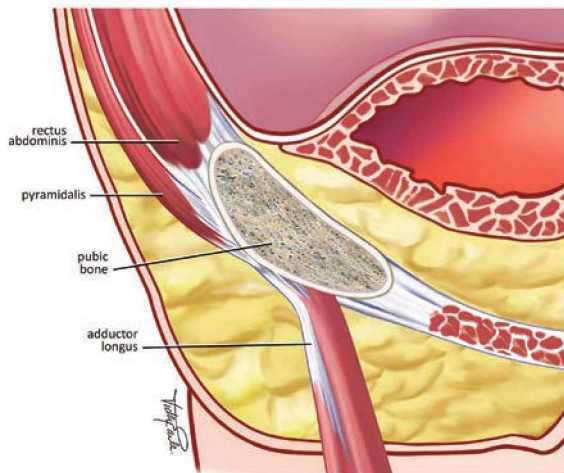
Pubic symphysis

The pubic symphysis is a secondary cartilaginous joint connecting the two pubic rami. It is supported superiorly by ligaments and inferiorly by the superior pubic ligament and the inferior or arcuate pubic ligament respectively.² The joint is lined with hyaline cartilage. A fibrocartilaginous disc separates the two pubic bones. The pubic symphysis is the site of numerous musculotendinous attachments, which act to dynamically stabilise the anterior pelvis and the transference of large

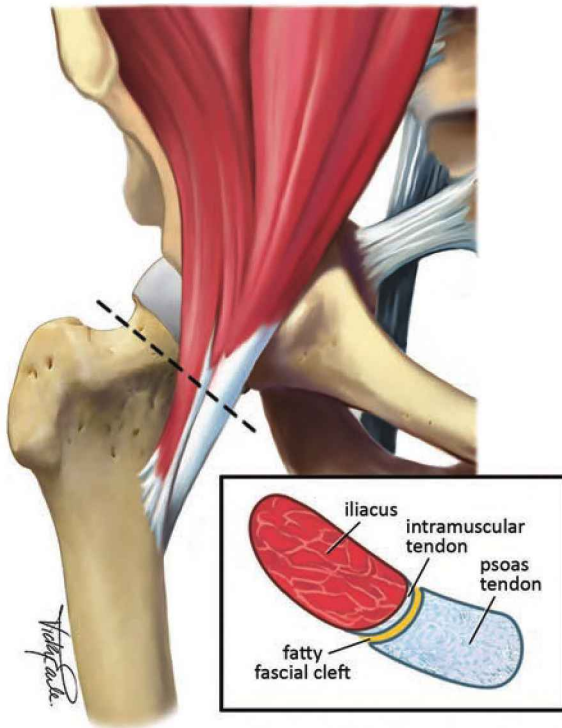


(a)

Figure 32.1 (a) Overview of the anatomy of the groin



(b)



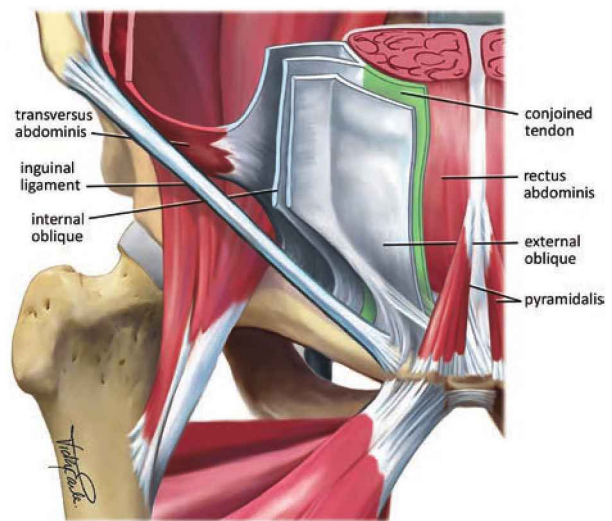
(c)

Figure 32.1 (cont.) (b) Anatomy of the aponeurosis on the anterior side of the pubic symphysis (c) Anatomy of distal iliopsoas tendon

forces acting across this joint, especially during specific sporting activities such as kicking and cutting.

Hip adductors

The hip adductors include pectineus, gracilis and adductor longus, brevis and magnus. The adductors are



(d)

Figure 32.1 (cont.) (d) Anatomy of the muscles in the inguinal region

the most commonly injured groin muscle in soccer^{3, 4} and the adductor longus is the most commonly injured adductor muscle.⁵ It inserts at the middle one-third of the linea aspera of the femur and has an interesting proximal origin. The origin is predominantly muscular (around 60%). The anterior tendinous portion continues across the front of the symphysis to fuse with the distal rectus and external oblique forming an aponeurosis.⁶ This aponeurosis has fibres that insert from the adductors and rectus directly into the pubic symphysis joint capsule and disc.² The adductor longus not only acts as an adductor but also as an important hip flexor when the hip is in an extended position, and therefore is also highly involved in kicking.⁷

The anatomy of the aponeurosis on the anterior side of the pubic symphysis is shown in Figure 32.1b.

Hip flexors

The iliopsoas is the primary hip flexor but rectus femoris, sartorius, tensor fascia lata and a number of hip adductors also contribute to hip flexion. The iliopsoas is made up of the iliacus, which originates from the iliac fossa in the pelvis and the psoas muscle.⁸ The psoas arises from the vertebral bodies of T12 and L1-5 and the intervertebral discs. The two muscles converge distally to form the iliopsoas. Although the tendon is often depicted as a common tendon, the lateral portion of the iliacus extends distally to insert directly onto the femur and at the level of the hip joint there are distinct tendons.⁸

The distal iliopsoas tendon anatomy is shown in Figure 32.1c.

Inguinal region

The anatomy of the inguinal canal is complex and extensively described in detail in classic anatomical texts. The abdominal muscles—transversus abdominis, internal and external obliques and the rectus abdominis—are all intimately related in this region. The inguinal ligament is formed by a distal shelving of the external oblique and the conjoined tendon from a convergence of transversus abdominis and the internal oblique at the medial ends. The conjoined tendon also inserts into the lateral rectus sheath.⁹

The anatomy of the muscles in the inguinal region is shown in Figure 32.1d.

Summary of anatomy

The anatomy of the groin has clinical implications as it demonstrates a high degree of confluence of the muscle, tendon, bone and joint structures. This connectivity means that pain provocation or stress tests do not load a single anatomical structure in isolation. This may explain why patients who present with groin pain often have diffuse, poorly localised pain and multiple clinical findings.

EPIDEMIOLOGY

Groin pain and injury is common with sports that involve kicking, rapid acceleration and deceleration, and sudden change of direction.^{10, 11} Until recently, there was no agreement on terminology, definitions or classification of groin pain in athletes.¹² This makes comparing epidemiological studies difficult for two main reasons. Firstly, multiple injury definitions and classifications systems have been used; some combined hip and groin, some classify groin separately, and almost all use different terminology for various groin injuries, which are often not specifically described.¹³ Additionally, the nature of injury onset is often poorly reported (e.g. acute versus chronic; traumatic versus overuse).

The second difficulty is that groin injury is often only recorded when an athlete misses a training session or match. Therefore, the prevalence of groin injury may be underestimated as those athletes who continue to train and play through injury often are not recorded.

Incidence—soccer

A recent systematic review on the incidence of groin pain in soccer included 34 articles.¹⁴ In general, there was risk of bias in the studies due to participant selection (18 studies), exposure (17 studies) and precision of the estimates given (18 studies).

The main findings of the review were:

- groin injuries in male club soccer accounted for 4–19% of all injuries
- groin injuries in female club soccer accounted for 2–14% of all injuries

- analysis of 29 studies found a higher proportion of groin injuries in men (12.8%) than women (6.9% with an absolute difference of 5.9%, 95% with a confidence interval of 4.6–7.1%)
- groin injury rates in males were 0.2–2.1/1000 hours and 0.1–0.6/1000 hours in women (rate ratio 2.4, 95% with a confidence interval of 2.0–2.9).

In adductor-related injuries in elite male soccer the re-injury rate has been reported to be 15%.³ The recovery period for a re-injury in elite soccer players is almost twice as long compared to the original injury emphasising the importance of avoiding re-injury by finishing treatment properly the first time.³

Incidence—elite sports other than soccer

A recent review identified 31 papers where more than 10 team seasons of groin or groin region injury incidence were reported.¹¹ These studies used varying injury definitions and also considered varying injury categories from general to specific (all groin/hip region injuries, groin injuries, adductor muscle strains, intra-articular hip injuries). The main findings were as follows.

- When playing the same sport, males had greater injury incidence of groin injury than females (RR 2.45, 95% with a confidence interval of 2.06–2.92).
- The sports with a high rate of groin injury were ice hockey and the football codes where field kicking is common (Australian rules football, Gaelic football).
- Within the football codes, player positions involving more kicking had a higher incidence.
- Uniform injury classification and definitions would improve comparison of results across studies.

Prevalence

As mentioned previously, using only time loss or medical attention definitions to count athletes as being injured probably misses many less-severe injuries. The injuries leading to time loss are thought to represent the tip of the iceberg. A study on football players showed that 49% had experienced groin pain in the previous season and that this had lasted longer than 6 weeks in 31%.¹⁵

Distribution of acute injuries

Few studies have closely examined which proportion of groin injuries has an acute onset. In a Danish study of 998 soccer players it was found that on the 58 injuries registered, 20 had acute onset, 31 were classified as overuse and 7 could not be classified.⁴ This would suggest that acute onset injuries represent a significant proportion of the total injuries in soccer players. Defined terminology using specific classification systems for

injury distribution have not been used in many studies. The study of 998 soccer players used a terminology and classification system almost identical to the one we use in this chapter (see section below) to classify the 58 injuries.⁴ Of the 58 injuries, 16 (27%) could not be classified and 24% had multiple entities. Adductor-related groin pain was the most frequent injury (51%) with iliopsoas (30%) and abdominal (19%) also being diagnosed frequently.

A recent study examined acute onset groin injuries in 110 athletes presenting to a sports medicine hospital within 7 days of injury.⁵ The majority (76%) were football code players with 10% basketball and 8% handball players. In the football codes 40% of the injuries occurred during kicking, where change of direction was the most frequent activity (31%) during onset in other sports. Clinical examination and imaging (MRI or ultrasound) were performed in all 110 athletes. The majority of injuries affected the adductors (66%), with iliopsoas (17–25%) and rectus femoris (15–23%) injuries also being frequent.

Other case series have examined the distribution of groin pain in those presenting to different clinicians in multiple countries. These series found adductor-related groin pain,¹⁶ inguinal-related groin pain¹⁷ and hip-related groin pain¹⁸ to be the most common problems. As these studies all suffer from referral bias, they are less helpful than studies where a single cohort is followed over time. They do, however, all report multiple problems occurring in a significant number of those who present.

PRACTICE PEARL

Adductor-related groin pain is the most common groin injury in the football codes and ice hockey.

RISK FACTORS

A lack of agreement on definitions and terminology of groin pain and the various definitions of when injury is present make the literature on risk factors challenging to understand. A previous comprehensive review¹⁹ was updated in 2015.²⁰ This systematic review covered 29 studies and summarised literature with a levels-of-evidence approach. The majority of studies were prospective cohort studies. The review found that there was level 1 and 2 evidence that the following are associated with an increased risk of groin injury in athletes:

- previous groin injury (level 1)
- higher level of play (level 1)
- reduced hip adduction (absolute and relative to abduction) strength (level 2)
- lower levels of sport-specific training (level 2).

Whittaker et al. also highlighted that to date there has been virtually no investigation of the relationship between exposure/athletic load and risk. Future studies should be large enough to ensure around 50 injury cases to examine for moderate to strong associations and include clear injury definitions.²⁰

Where risk factor studies should ideally be prospective in design, cross-sectional and case control studies allow multiple factors to be examined in smaller populations. While no prospective relationship can be determined, these factors can be considered in future studies on risk factors, investigations and treatment. A systematic review with a meta-analysis examining these cross-sectional factors differentiating athletes with and without groin pain was carried out in 2015.²¹ Seventeen cross-sectional studies were included, of which 10 were high quality. In total, 62 different measures were investigated. Eight studies were suitable for meta-analysis. Meta-analysis showed that athletes with hip and groin pain had:

- pain and lower strength on adductor squeeze test
- reduced hip internal rotation and bent knee fall out, but that hip external rotation was the same as controls.

For other factors, a levels-of-evidence synthesis was performed. There was strong evidence that athletes with hip and groin pain have an association with:

- lower patient-reported outcome (PROM) scores
- altered trunk muscle function.

TERMINOLOGY AND DEFINITIONS

A systematic review on the treatment of groin pain in athletes included 72 studies, in which 33 different diagnostic terms were used.¹³ The 'Doha agreement meeting on terminology and definitions in groin pain in athletes' was convened to attempt to resolve this problem.²² The aim was to agree on a standard terminology, along with accompanying definitions. A Delphi process was used to inform agreement. Unanimous agreement on the following terms and definitions was reached. Some considerations are presented here as a preamble to the terms.

PRACTICE PEARL

The preferred umbrella term was 'groin pain in athletes'. This was favoured over others (e.g. athletic pubalgia, athletic groin pain, sports groin pain, athletes' groin) because it is clearly descriptive and cannot be misunderstood to be a diagnostic term.

A number of previously popular terms were considered and rejected during the meeting. The terms that the group chose not to recommend were: adductor and

iliopsoas tendinitis or tendinopathy, athletic groin pain, athletic pubalgia, biomechanical groin overload, Gilmore's groin, groin disruption, hockey-goalie syndrome, hockey groin, osteitis pubis, sports groin, sportsman's groin, sports hernia and sportsman's hernia. The term 'entity' was chosen to reflect the recognisable pattern of symptoms and signs exhibited by the athlete.

To ensure generalisability and straightforward use in everyday practice, a clinically based classification system is recommended. This means that a thorough history and physical examination are essential. In the history, in all cases the athlete should report pain in the affected region that worsens on exercise. Palpation is important to identify the painful structures and must be precise, as numerous structures in the groin region are in close proximity and can refer pain to overlapping areas. The term 'tenderness' is defined in this system as discomfort or pain when the area is touched and that the athlete recognises this to be their specific injury pain.

Pain on resistance testing should also be felt in the affected structure. For example, in adductor-related groin pain, the pain on resisted adduction testing should reproduce the athlete's recognisable pain in the adductors. Pain felt in a different location (e.g. the inguinal region on resisted adduction testing) would not signify adductor-related groin pain. The techniques of physical examination are covered in the 'Clinical overview' section.

Classification

Acute groin injuries

Acute injuries refer to the manner in which the athlete first felt the pain (i.e. sudden onset). In general, the system proposed in this section could potentially be used to classify the majority of acute groin injuries into entities. There is far less literature on acute groin injuries when compared to longstanding groin pain.

Consequently, a careful history—along with examination comprising palpation, resistance testing and stretching—is critical. In cases with severe pain, it may be hard to perform a thorough physical examination. It should be recognised that some acute groin injuries such as proximal rectus femoris injuries do not readily fit into the current Doha classification system. Further work will be needed to refine this system to fully encompass the entire spectrum of acute groin injuries.

Longstanding groin pain

The exact duration considered to be longstanding is not further defined. Longstanding groin pain can start either gradually or suddenly and does not refer to the mechanism of onset, but only to the duration of symptoms.

The classification system has three major subheadings of groin pain in athletes.

1. Defined clinical entities for groin pain:

- adductor related
- iliopsoas related
- inguinal related
- pubic related.

The locations of the four entities for groin pain are shown in Figure 32.2.

2. Hip-related groin pain. Pain from the hip joint (Chapter 31) should always be considered as a possible cause of groin pain.

3. Other conditions causing groin pain in athletes. Besides the defined clinical entities and the hip, there are many other possible causes for groin pain in athletes. A high index of clinical suspicion is needed to identify these and clinicians need to be alert to the possibilities, especially when the complaints cannot easily be classified into one of the common defined

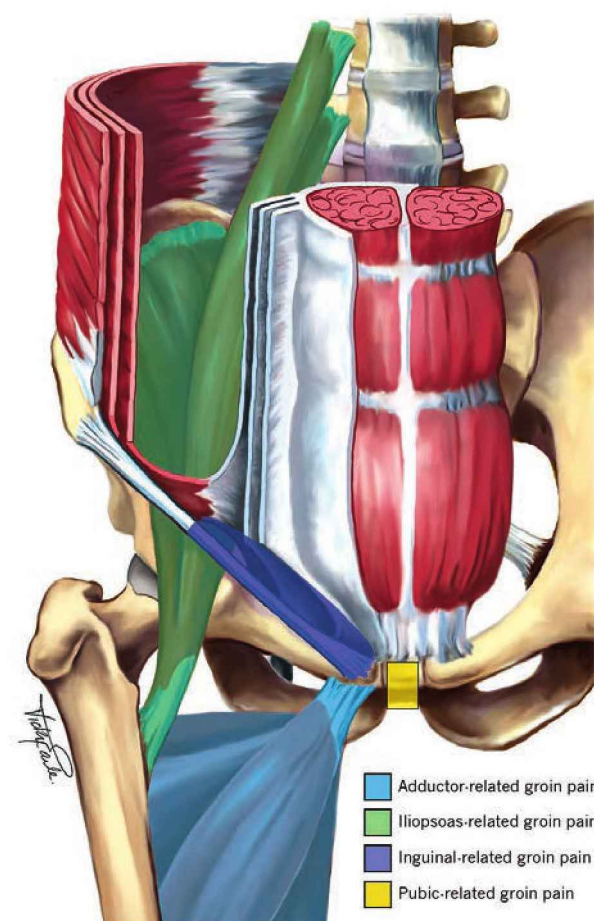


Figure 32.2 Defined clinical entities for groin pain

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Table 32.1 Overview of some possible causes of groin pain in athletes

Entities defined during Doha agreement meeting	Other musculoskeletal causes	Not to be missed
Adductor-related groin pain Iliopsoas-related groin pain Inguinal-related groin pain Pubic-related groin pain Hip-related groin pain	Inguinal or femoral hernia Post-hernioplasty pain Nerve entrapment: – obturator – ilioinguinal – genitofemoral – iliohypogastric Referred pain – lumbar spine – sacroiliac joint Apophysitis or avulsion fracture – anterior superior iliac spine – anterior inferior iliac spine – pubic bone	Stress fracture – neck of femur – pubic ramus – acetabulum Hip joint – slipped capital femoral epiphysis (adolescents) – Perthes' disease (children and adolescents) – avascular necrosis /transient osteoporosis of the head of the femur – arthritis of the hip joint (reactive or infectious) Inguinal lymphadenopathy Intra-abdominal abnormality – prostatitis – urinary tract infections – kidney stone – appendicitis – diverticulitis Gynaecological conditions Spondyloarthropathies – ankylosing spondylitis Tumours – testicular tumours – bone tumours – prostate cancer – urinary tract cancer – digestive tract cancer – soft tissue tumours

clinical entities. An unclear relationship between the pain and loading during sport, nocturnal or extremely severe pain or a lack of response to treatment should trigger more suspicion. There are numerous possible causes, a number of which are listed in Table 32.1. The main categories are orthopaedic, neurological, rheumatological, urological, gastrointestinal, dermatological, oncological and surgical; however, this list is not exhaustive as many rare conditions could possibly cause pain in the groin region.

A careful history and physical examination covering more than only the musculoskeletal system, and appropriate additional investigations or referrals are critical for identifying other possible causes. Some of these conditions are discussed in more detail at the end of this chapter.

CLINICAL OVERVIEW

As previously alluded to, it is important to appreciate the anatomy of the hip and groin, and undertake a careful history and examination. The clinical approach can be difficult as the anatomy around this region is complex and often multiple pathologies coexist. Pain may be difficult to localise and be accompanied by vague symptoms. An insidious onset with varying locations of the pain may also cloud the clinical presentation.

History

The athlete experiences groin pain, which is usually located in the medial upper thigh around the proximal adductors and pubic rami. The pain can also be more central in the upper thigh inferior to the inguinal ligament around the hip joint/distal iliopsoas region. The pain frequently starts

in one region and is unilateral, but it can gradually spread to other regions and become bilateral. Pain superior to the inguinal ligament can be felt in the region of the inguinal canal. Groin pain in athletes is aggravated by exercise, with running, twisting/turning and kicking being the most challenging activities. The athlete and coach usually notice a decrease in sports performance, especially related to performing explosive sporting actions such as kicking, accelerating/decelerating, turning and so on.

Pain pattern

The onset of groin pain in athletes can be acute or gradual, but with both types of onset, groin pain can become longstanding. In the early stages, the longstanding groin pain patient typically presents late during the physical activity or after activity, with pain and stiffness next morning. The pain and stiffness then gradually lessen with daily activities and warming-up for the next training session or match. When the condition worsens, pain is present immediately upon exercise. Non-steroidal anti-inflammatory drugs (NSAIDs) tend to decrease pain, but will usually not result in a lasting cure. Short periods of rest reduce the severity of the symptoms, but on resumption of sporting activities the pain often returns to its original intensity and severity. The natural history is one of progressive deterioration with continued activity until symptoms prevent participation in the sporting activity.

Where is the pain located?

The localisation of the pain is important in determining which structure may be causing the pain. Adductor-related groin pain is often located at the attachment of the adductor longus tendon to the pubic bone. Iliopsoas-related groin pain is located more centrally in the groin and proximal thigh. The type of activity that aggravates the pain gives a clue to the primary site of the problem. Side-to-side movements, kicking, twisting and turning activities which aggravate the pain suggest adductor-related groin pain. Straight-line running or jogging suggests iliopsoas-related groin pain. Pain with sit-ups and/or coughing, may suggest an inguinal-related groin pain. Note that these clinical observations are guidelines rather than hard and fast rules, as many of the symptoms overlap.

Pain that becomes progressively worse with exercise may suggest a stress fracture or an apophysitis in young athletes. A history of associated pain such as low back or buttock pain indicates that the groin pain may be referred from another site (such as the hip, the sacroiliac joint or the spine). A full training history should be taken to determine if any recent changes in training (e.g. a generalised increase in volume or intensity, the introduction of a new exercise or an increase in a particular component of training) may have led to the development of the groin pain.

Clinical examination

Clinical examination includes the adductor muscles, the pelvic bones, the hip joint, the hip flexors (including tensor fascia lata and sartorius) and the lower abdominal muscles. The lumbar spine and sacroiliac joints should also be examined. Pelvic alignment must be assessed and large leg length discrepancies (>2 cm) noted.

Examination involves the following.

1. Observation
 - a. standing
 - b. walking (Fig. 32.3a)
 - c. squatting, running, jumping
2. Active movements
 - a. hip flexion/extension (Chapter 31)
 - b. hip abduction/adduction
 - c. hip internal/external rotation
 - d. lumbar spine movements (Chapter 29)
3. Passive movements
 - a. passive movement–adductor muscle stretch (Fig. 32.3b)
 - b. modified Thomas Test (Fig. 32.3c)
4. Resisted movements
 - a. resisted movement–single leg (Fig. 32.3d)
 - b. resisted movement–squeeze test
 - i. with hips in 45° flexion (Fig. 32.3e-i)
 - ii. with hips in neutral (Fig. 32.3e-ii)
 - c. hip flexion (Fig. 32.3f)
 - d. abdominal muscles (Fig. 32.3g)
5. Palpation
 - a. adductor (Fig. 32.3h)
 - b. psoas above inguinal ligament (Fig. 32.3i)
 - c. psoas below inguinal ligament (Fig. 32.3j)
 - d. inguinal region–external inguinal ring (Fig. 32.3k)
 - e. inguinal region–scrotal invagination (Fig. 32.3l)
 - f. pubic symphysis (Fig. 32.3m)
6. Hip-related groin pain
 - a. examination of the hip joint (Chapter 31)
7. Screening for other pathology
 - a. stress fractures–neck of femur and pubic ramus
 - b. thoracic spine (Chapter 28) and lumbar (Chapter 29)
 - c. sacroiliac joint (Chapter 30).

Assessment of severity

While clinical examination is essential for diagnosing and classifying groin pain in athletes, it gives little objective information on the severity of the condition. The severity, insights into the impairments, activity limitations and accompanying participation restrictions can be measured with strength, range of motion and patient-reported outcome tools.

Strength

The assessment of muscle strength provides better understanding of the degree of impairment in athletes



(a)



(c)



(b)



(d)

Figure 32.3 Examination of the patient with longstanding groin pain (a) Walking—assess lower limb alignment from in front, particularly for evidence of excessive internal or external hip rotation and muscle wasting (b) Adductor muscle stretch—tightness or the presence of pain (c) Passive stretching of the hip flexors for tightness or the presence of pain. Following stretching, the athlete can push upwards with the leg to assess hip flexor strength in outer range and for the presence of pain during strength testing (d) Resisted movement—adductors in outer range—for strength and the presence of pain



(e)-(i)



(f)



(e)-(ii)



(g)



(h)

Figure 32.3 (cont.) (e) (i) Squeeze test—resistance testing of the adductors with the hip in 45° flexion (ii) Squeeze test—resistance testing of the adductors with the hip in neutral and long lever for strength and the presence of pain (f) Hip flexors with the hip in 90° flexion—for strength and the presence of pain (g) Resistance testing of the abdominals for strength and the presence of pain (h) Palpation of the proximal adductors for tenderness



(i)



(ii)



(j)



(m)



(k)

Figure 32.3 (cont.) (i) Abdominal portion of the psoas for tenderness (j) Palpation of the distal iliopsoas for tenderness (k) Transabdominal palpation of the external inguinal ring for tenderness (l) Scrotal invagination for palpation of the inguinal ring and canal for tenderness and/or hernia (m) Palpation of the pubic symphysis for tenderness

with groin pain. It can be quantified objectively, and is a great help in monitoring clinical progress or deterioration. Adductor strength can be reliably assessed using handheld dynamometry and quantified either unilaterally (Fig. 32.4a) or as a squeeze test (Fig. 32.4b and c) and is especially relevant in athletes with adductor-related

or pubic-related groin pain. It may also be relevant to include abductor strength (Fig. 32.4d) before return to sport in athletes with adductor-related groin pain or acute injury, as an eccentric hip adductor/abductor ratio of <0.8 appears to increase the risk of future adductor injuries.

For isometric and eccentric measures, the minimal detectable change when using the same tester is around 10–15%, while being around 20% for the isometric adduction/abduction (ADD/ABD) ratio.²³ While eccentric ratios for ice hockey players²⁴ and isometric ratios for soccer players seem to be around 1.0,²⁵ these ratios may be sport-specific. Clinically, this means that reductions in strength of 15% can be measured at the individual level. This may indicate that athletes either have, or are



(a)



(c)



(b)



(d)

Figure 32.4 Hand-held dynamometry provides an objective measure of strength (a) Unilateral adductor (b) Bilateral (squeeze) in hip and knee flexion (c) Squeeze in hip abduction (d) Abductor

at increased risk, of developing groin pain. Studies in European and Australian soccer players and Gaelic football players with groin pain have shown that strength deficits of around 20% or more seem to exist in eccentric strength and during the squeeze test.²⁶⁻²⁹

Range of motion

Passive hip joint internal rotation range of motion in degrees can also be reliably assessed using a goniometer or inclinometer, although severe restrictions in passive range of motion are more closely related to hip intra-articular conditions, and less often to groin pain related to tendon attachments and the pubic bone. Assessment of the hip joint is demonstrated in Chapter 31.

Patient-reported outcome measures (PROMs)

The use of validated outcome measures should also be encouraged. In a systematic review from 2015 on the clinimetric properties of PROMs for young physically active individuals with hip and/or groin pain, the

Copenhagen Hip and Groin Outcome Score (HAGOS) was recommended.³⁰ HAGOS is reliable, valid and responsive in athletes with hip and/or groin pain, and therefore not restricted to any specific pathology or entity. Reference values for soccer players have been provided.³¹ It measures six relevant subsets related to groin pain in athletes, including pain, symptoms, activities of daily living, participation in physical activity, sports function and quality of life. Reference values have been given for soccer players without groin pain, and while their average and upper boundary is 100 points, the lower boundary goes down to 70 points in the sporting function and participation scores.³² On average, Gaelic footballers and soccer players with adductor-related groin pain display scores in the area of 50 points and below when having had pain for at least 4 weeks.^{26, 27}

Imaging

The current evidence for the use of radiographs, MRI and ultrasonography in groin pain is based on relatively

Outcome measures

Physical assessments should include some of the following valid and reliable measures according to relevance to the clinical findings. The following measures are of relevance.

Strength (hand-held dynamometer (HHD) or sphygmomanometer)

- hip adduction strength—squeeze, isometric, eccentric (Fig. 32.4(a)–(c))
- hip abduction strength—squeezing, isometric, eccentric ADD/ABD ratio (Fig. 32.4(d))
- hip flexion strength—squeezing, isometric, eccentric

Range of motion (goniometer or inclinometer)

- hip internal rotation
- hip external rotation

- hip abduction
- hip extension (Thomas test position)

Patient-reported outcome measures (PROM)

- Copenhagen Hip and Groin Outcome Score, six subscales
 1. Pain (0–100)
 2. Symptoms (0–100)
 3. Activities of daily living (0–100)
 4. Sport and recreation (0–100)
 5. Participation in physical activity (0–100)
 6. Quality of life (0–100).

few heterogeneous studies which are of varying methodological quality.³³ The correlation between clinical findings, athletes' symptoms and the identified radiological abnormalities is quite weak.

Two recent studies have illustrated this.^{34, 35} Branci et al. compared the standardised MRI findings of three groups of male adult athletes matched for level of sports activity.³⁴ Positive MRI findings were significantly more frequent in soccer players compared with non-soccer players irrespective of whether or not the soccer players had symptoms, suggesting that these MRI changes may be associated with soccer play itself rather than clinical symptoms. In a 4-year prospective cohort study of young soccer academy players using serial clinical and imaging assessments, Robinson et al. found a high prevalence of positive imaging findings, including pubic bone marrow oedema (BMO), in asymptomatic soccer players as well as in those with injury.³⁵ These two studies show that imaging should not be used in isolation when diagnosing groin pain in athletes.

Radiography

Plain radiographs of the pelvis are commonly recommended in many cases to evaluate the hip joints and the pubic bones. If the patient history and the clinical examination suggest that the groin pain could be hip-related, an anteroposterior pelvic radiograph and a true lateral radiograph should be obtained. The α angle, the centre-edge (CE) angle and other relevant measurements can be examined, looking for any bony morphology that can result in femoroacetabular impingement (FAI).

FAI morphology has been found to coexist with longstanding adductor-related groin pain and the presence of FAI morphology is not necessarily a sign of hip joint

pathology (Chapter 31).³⁶ In this study of athletes with longstanding adductor-related groin pain, 94% had at least one radiographic sign of FAI on the X-ray. In a different cohort of athletes with longstanding adductor-related groin pain who also had plain radiographs performed, follow-up was done after 8–12 years. The study found no evidence that bony hip morphology related to FAI or dysplasia prevented successful outcome of an exercise treatment for this musculotendinous injury.³⁷

Plain radiograph of the pubic symphysis often shows osteolytic changes, irregular widening and sclerosis along the rami of the os pubis in soccer players. Historically these changes were interpreted as a diagnosis using the term 'osteitis pubis' (Fig.32.5a). These changes can also be seen before radiographic changes are present as pubic BMO on MRI. As described earlier, this is not that specific for injury, but probably reflects the considerable strain that the pelvic girdle is exposed to in the kicking sports in particular.³⁴

It is likely that there is a continuum of bone stress and that there is not a strong relationship between the degree of stress and the pain felt by an athlete. This was alluded to by Harris et al. in 1974,³⁸ and the case control study of Branci et al.³⁴ found that higher grades of pubic BMO, along with protrusion of the disc in the symphyseal joint, were associated, but not strongly, with pain compared to soccer-playing controls. A bone biopsy study of athletes with pubic-related groin pain found evidence of new woven bone formation but a lack of inflammation, and so using the term 'osteitis pubis' cannot be recommended.³⁹

In the skeletally immature adolescent athlete, plain radiographs are used to detect osseous avulsions in musculotendinous distraction injuries and epiphysiolysis of the growth plate of the femoral neck. The plain radiograph is also a good primary examination for

neoplasms of the bone. Even in seemingly healthy athletes, this should be considered as a possible cause of unexplained groin pain or when there is an unusual response to treatment. Radiographs are often used as a first-line investigation if a pelvic stress fracture is suspected; if a stress fracture is visible, radiographs can confirm the clinical suspicion. Clinicians should note that radiographs are frequently negative when stress fractures are present and cannot be used to rule out stress fractures.

Magnetic resonance imaging

Magnetic resonance imaging (MRI) is a very sensitive, but not always equally specific, imaging technique. It is used widely, but the results should be interpreted with caution.³³ The high sensitivity means that when imaging athletes with MRI many 'abnormalities' will be seen, but that the clinical relevance of these findings is doubtful in terms of making a diagnosis or estimating prognosis. Ongoing studies are examining the value of MRI in the diagnosis of acute groin injuries and the results are promising. Some common MRI findings are pubic BMO, adductor tendon changes and findings around the aponeurosis anterior to the symphysis.

Pubic BMO is a common finding. Studies have shown that the percentage of athletes without groin pain found to have BMO on MRI is high (31–61%).^{34, 40, 41} To date, no studies have examined the effect of the presence/absence or grade of BMO on the time to recovery.

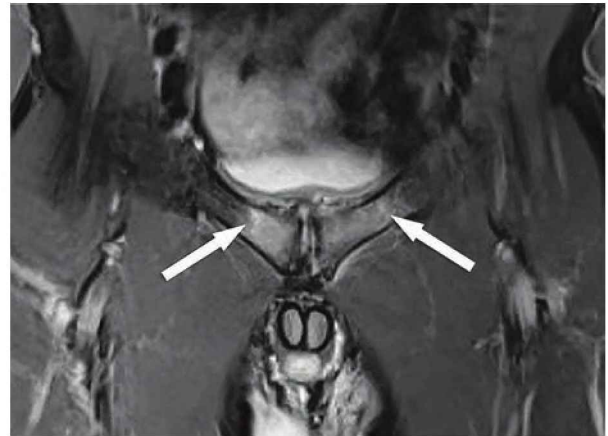
Changes in MRI signal of the adductor tendon around the proximal insertion have been referred to as 'enthesopathy',⁴² 'tendinopathy'³⁴ and 'pubic aponeurosis defect'.⁴³ This demonstrates the confusion with regard to describing these virtually identical radiological findings of increased signal intensity in the proximal adductor tendon. Some have found the reliability of determining if adductor tendon signal changes are present to be poor,⁴⁴ while others found it to be good.³⁵

The differing terminology may explain why some studies found changes in the tendon to be present in more than 70% of asymptomatic players,³⁴ while others in only 11%.⁴¹ As with pubic BMO, there have been no studies to date examining whether athletes who have adductor tendon signal changes take longer to recover than those without. In other tendons, only weak associations have been found between structure and symptoms, and a normalisation of tendon structure after treatment was not associated with better outcome.⁴⁵

The secondary cleft sign (Fig. 32.5c) was first described as being present when contrast medium injected into the pubic symphysis extended beyond the midline or inferior to the joint.⁴⁶ Other groups used a high signal intensity line extending laterally and inferiorly from the



(a)



(b)



(c)

Figure 32.5 (a) X-ray showing characteristic moth-eaten appearance consistent with longstanding pubic bone stress (b) MRI BMO. Arrows point to BMO in pubic bones on both sides of symphyseal joint (seen as diffuse increased signal intensity on fluid sensitive sequences) (c) MRI secondary cleft. Hyperintense cleft (arrow) on fluid-sensitive sequences, extending inferiorly and laterally to the symphyseal joint



(d)

Figure 32.5 (cont.) (d) CT scans demonstrating stress-related changes at the pubis: (A) Cystic changes; (B) Asymmetrical irregularities and cystic changes; and (C) Unilateral widening of the pubic apophysis

inferior part of the symphysis of fluid sensitive sequences, without the use of contrast, to define the secondary cleft.³⁴ The secondary cleft was initially found to be present in two-thirds of athletes with groin pain and absent in 70 control subjects who were rowers.⁴⁶ Later studies found it to be present at similar prevalences in soccer players with and without groin pain,³⁴ and in a prospective study of young asymptomatic soccer players it did not predict the onset of groin pain.³⁵ Again there is a lack of data on how the presence of a secondary cleft on MRI may affect the prognosis.

PRACTICE PEARL

In summary, the common MRI findings of pubic BMO, adductor tendon changes and secondary cleft are found frequently in healthy players, and in general do not discriminate well between those with and those without pain.

There is a lack of data on how these findings should influence treatment decisions and their meaning for helping make a more accurate prognostic prediction.

Stress fractures, neoplasms and other occult bony injuries are rare and when suspected, but not visible on radiography, MRI is the best investigation. Even in seemingly healthy athletes, these should be considered as a possible cause of unexplained groin pain.

Ultrasonography

Ultrasonography (US) has gained increasing popularity in recent years (Chapter 15). Scientific evidence for its use in groin injuries is very limited.³³ It has shown promise with other muscle and tendon injuries.^{47, 48} There is a significant discrepancy between the widespread clinical use and experience gained in many countries, and the amount of data available in the literature to help clinicians improve interpretation of findings.

Computed tomography scan

Computed tomography (CT) scan offers excellent visualisation of the pubic symphysis joint, but requires high doses of radiation. Its general use in the young athletic population cannot be advised due to the significant radiation risk and a lack of data on normal findings in the athletic population. An interesting recent paper which examined CT scans of young males who had undergone CT scans for reasons other than groin pain showed that the pubic apophysis closes in the late teens and early twenties (Fig. 32.5d).⁴⁹ While the apophysis can be visualised well on CT, it can also be clearly seen on MRI, which is safer and therefore the preferable modality.

Radioisotopic bone scan

Given the high radiation dose of bone scans, and the fact that MRI allows good visualisation of both bony and soft tissue structures in the pelvis, the use of bone scan as a routine investigation for groin pain in athletes cannot be recommended.

ACUTE GROIN INJURIES

An acute groin injury usually involves one or more musculotendinous structures.⁵ The lesion can be in the musculotendinous junction, but in some cases the tendon itself or the entheses where the tendon inserts into the bone is the site of the injury. These injuries usually happen during explosive actions such as kicking, reaching with the leg, sudden change of direction or other movements where the muscle is being stretched during forceful contraction.⁵

Diagnosis

The most common acute injury in the groin is to the adductors, especially the adductor longus, which has been verified both clinically and with diagnostic imaging.⁵ Good agreement between clinical diagnosis and diagnostic imaging seems to exist for acute adductor longus injuries.⁵

However, acute injuries to other musculotendinous structures such as the iliopsoas, proximal rectus femoris or the inguinal canal/conjoined tendon are more difficult to localise clinically in the acute stage. Acute injuries to iliopsoas and rectus femoris are not infrequent and should be considered relevant differential diagnoses that may need imaging to be correctly identified.

In the acute stage, ultrasound has shown clinical promise as a simple method for localising specific structures in the groin, due to the visualisation of oedema or structural derangement. MRI has advantages when considering the underlying symphysis joint and adjacent bone, thus allowing for more complete assessment, and may be less operator dependant. Diagnostic imaging may therefore play an important differential diagnostic role when examining acute groin injuries, but may also have a role in guiding treatment choices from a very early stage in relation to which structures and what kind of tissue should be targeted in the rehabilitation of the particular injury. At the present time, however, there is no evidence that the use of imaging does lead to better clinical outcomes or more accurate prediction of the duration of injury.

If the initial injury is not treated appropriately in the first place or if a player is returned to sport too quickly, it may develop into a more longstanding injury. If the athlete does not get appropriate treatment and continues to participate, the injury pattern may start to resemble the more overuse type where more and more sport-specific activities become problematic. This type of injury can then take months to recover from and what started out as minor acute injury can develop into a longstanding problem.

The actual percentage of acute groin injuries that progress to longstanding groin pain in athletes is unknown. There is currently limited data on the recovery times for acute groin injuries.

LONGSTANDING GROIN PAIN

The Doha agreement confirms our approach to the classification of longstanding groin pain into the four clinical entities: adductor-related groin pain, iliopsoas-related groin pain, inguinal-related groin pain and pubic-related groin pain.²² As demonstrated in the 'Epidemiology' section an athlete can have more than one entity, in which case multiple entities can be diagnosed. The criteria for the diagnosis of each clinical entity are highlighted in each of the following sections in *italics*.²²

Adductor-related groin pain

Adductor-related groin pain is the most common acute and longstanding groin injury.³⁻⁵ Acute groin injuries are described elsewhere in this chapter and this section focuses on longstanding groin pain.

The patient with longstanding adductor-related groin pain usually complains of pain medially in the groin, most pronounced around the insertion of the adductor longus tendon at the pubic bone. The pain may radiate distally along the medial thigh. Pain during sprinting, cutting changes of direction and kicking the ball are common complaints.

Diagnostic criteria

The diagnostic criteria for adductor-related groin pain are *adductor tenderness* (Fig. 32.3h) AND *pain on resisted adduction testing* (Fig. 32.3d).²²

Evidence for treatment of groin pain in athletes

A systematic review from 2015 on the treatment of groin pain in athletes identified 72 studies.¹³ All studies were assessed for their methodological quality with a scoring system. The majority of studies (90%) were case series, of which 80% were retrospective. Only four studies were high quality. Twenty-five per cent reported on conservative treatment and 75% on surgical treatment for groin pain. A significant association was found with methodologically weaker studies reporting higher treatment success percentages. There has been no significant improvement in the quality of studies published over the last 30 years. A level-of-evidence approach was used to synthesise the results.

There is moderate evidence that for longstanding adductor-related groin pain:

- supervised active physical training results in a higher success and percentage of athletes returning to play than passive physical therapy modalities
- multi-modal treatment including manual adductor manipulation can result in a faster return to play, but not a higher treatment success, than a partially supervised active physical training program
- partial release of the adductor longus tendon seems effective for return to sport over time.

Additionally, there is moderate evidence that, for athletes with inguinal-related groin pain, laparoscopic hernia repair results in lower pain and a higher percentage returning to play than conservative treatment.

Treatment

The treatment of both acute and longstanding adductor-related groin pain will be considered together. The basis of the treatment of these conditions is exercise therapy and is based on Professor Per Hölmich's seminal paper published in *The Lancet* in 1999 which was the first randomised controlled trial (RCT) to examine treatment regimes in adductor-related groin pain.⁵⁰ The original exercise program is shown in full in the box below.

The exercise program comprises two modules. The first module consists of specific isometric and dynamic exercises to reactivate the adductor muscles and probably also modulates the pain.⁵¹ The athlete can have difficulties activating these muscles, probably as a result of negative feedback caused by the pain. The second module includes heavier resistance training as well as challenging balance and coordination exercises. The exercise program is performed three times a week, alternating with the exercises from module 1 on the days in between.⁵⁰

The total length of the exercise training period is between 8 and 12 weeks. Absence from training is

necessary as the functionality and strength of the adductor muscle group has to be re-established and pain brought under control without provocation by sport. Stationary cycling and general fitness training not involving the adductors can be allowed if pain-free.

Jogging is allowed after 6 weeks as long as it does not provoke any groin pain. When the patient is able to jog and undergo his or her regular treatment without pain, the treatment program is completed and the athlete is allowed to gradually progress to demanding sports-specific training followed by sports participation. No stretching of the adductor muscles should be done during this 2-3 month exercise training period.⁵⁰

Injection of corticosteroids at the adductor enthesis is not recommended, as there is no documented long-term beneficial effect and it might make it more difficult to monitor the training program needed to cure the groin pain. It is the experience of the authors that adductor tenotomy is extremely rarely indicated.

Many rehabilitation programs for adductor-related groin pain have used the Hölmich exercises as their basis.

Professor Per Hölmich's 1999 exercise program: an evidence-based foundation for contemporary rehabilitation programs

This program consists of static and dynamic exercises aimed at improving the muscles stabilising the pelvis and the hip joints, in particular the adductor muscles. The program consists of two parts.

- Module 1: Two-week familiarisation program—adductor activation.
- Module 2: More demanding exercises with heavier resistance training and balance and coordination. The training program is performed three times a week and the exercises from module 1 are performed on the days in between the treatment days. The total length of the training period is 8–12 weeks. Sport activities are not allowed in the treatment period. Pain-free bike riding is allowed. After 6 weeks, pain-free jogging is allowed. Return to sport is allowed when neither treatment nor jogging causes any pain.

Stretching of the adductor muscles is not advised, but stretching of the other lower extremity muscles, particularly the iliopsoas, is recommended.

Module 1: Static and dynamic exercises (2-week base training program)

Static

1. a. Adduction for 30 seconds against a soccer ball placed between the feet when lying in the supine position with the knees fully extended and the first toe pointing straight upwards.

1. b. Adduction for 30 seconds against a soccer ball placed between the knees when lying in the supine position with the knees and the hips flexed at 45° and the feet flat on the floor pointing straight ahead (Fig. 32.6a).

Exercises 1a and b should be repeated 10 times with 15-second recovery periods between each contraction. The force of the adduction should be just sufficient to reach the point where pain begins.

Dynamic

1. c. Sit-ups from the supine position with the hip and knee joints flexed at 45° and the feet against the floor. The sit-ups are performed as a straight abdominal curl and also with a quarter twist towards the opposite knee: five sets of 10 with 15-second recovery periods.
1. d. In the same starting position as for the sit-ups but clamping a soccer ball between the knees, the player does a combination of a sit-up while pulling the ball towards the head (Fig. 32.6b and c). The exercise is performed rhythmically and with accuracy to gain balance and coordination: five sets of 10 with 15-second recovery periods.
1. e. Wobble board training for 5 minutes.
1. f. Adductor lateral slide. Using a sliding board with an extremely smooth surface (or a very smooth floor) and wearing a low-friction sock on the sliding foot, one foot is positioned next to the sliding board and the other foot on

the board parallel to the first one. The foot on the board slides out laterally and is then pulled back to the starting position. The foot should be pressed against the surface through the whole exercise with as much force as can be tolerated within the patient's threshold of pain (Fig. 32.6d).

1. g. Forward slide. The same procedure is also done with the foot on the board placed in a 90° angle to the foot outside the board. Both exercises 1f and g are performed continuously for 1 minute with each leg in turn.

All the above exercises should be commenced carefully and the number of sets and range of motion gradually increased respecting pain and exhaustion.

Module 2: Dynamic exercises

This entire module is done twice at each training session for three training sessions per week with a day in between. *Note:* Module 1 is done on alternate days, so players are training a total of 6 days per week. Exercises 2a to e are done as five sets of 10 repetitions.

2. a. Lying on one side with the lower leg stretched and the upper leg bent and placed in front of the lower leg, the lower leg is moved up and down, pointing the heel upwards.
2. b. Lying on one side with the lower leg bent and the upper leg stretched, the upper leg is moved up and down, pointing the heel upwards.
2. c. Begin by standing at the end of a high couch and then lie prone so that the torso is supported by the couch. The hips are at the edge of the couch at 90° of flexion and the feet are on the floor. From this position, both hips are slowly extended so both legs are lifted to the greatest possible extension of hips and spine; legs are then lowered together.
2. d. Standing abduction/adduction using ankle pulleys/elastic bands. Begin with a low weight and gradually increase the weight but keep it submaximal (Fig. 32.7).
2. e. Standing on one leg, the knee of the supporting leg is flexed and extended rhythmically and in the same rhythm swinging both arms back and forth independently ('cross-country skiing on one leg') (Fig. 32.8). The non weight-bearing leg is not moved.



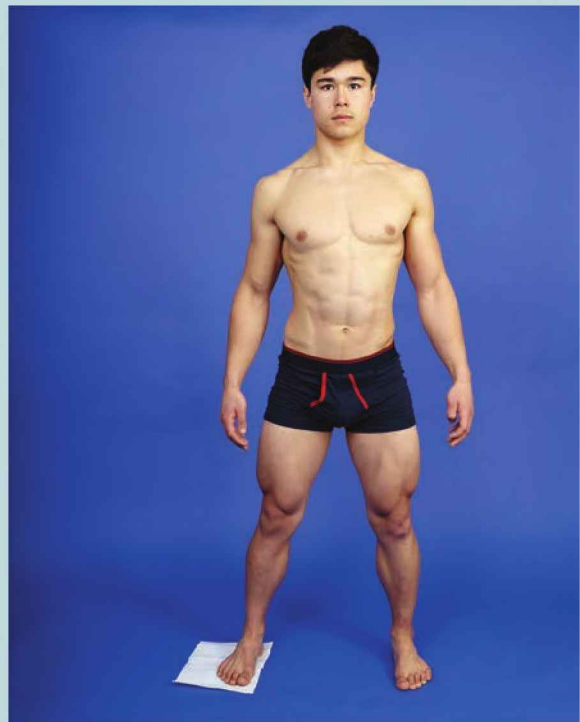
(a)



(b)



(c)



(d)

Figure 32.6 (a) Isometric exercise—adduction (b) and (c) Dynamic sit-ups for abdominals (d) Adductor lateral slide

continued



Figure 32.7 Standing hip adductor strengthening with elastic band. Dynamic strength and coordination exercise with emphasis on maximal concentric, isometric and eccentric strength of adductor muscles, with co-activation from contralateral

The balance and position is kept accurately and the exercise is stopped when this is no longer possible. Progression of the exercise is obtained by holding a 1 kg weight in each hand.

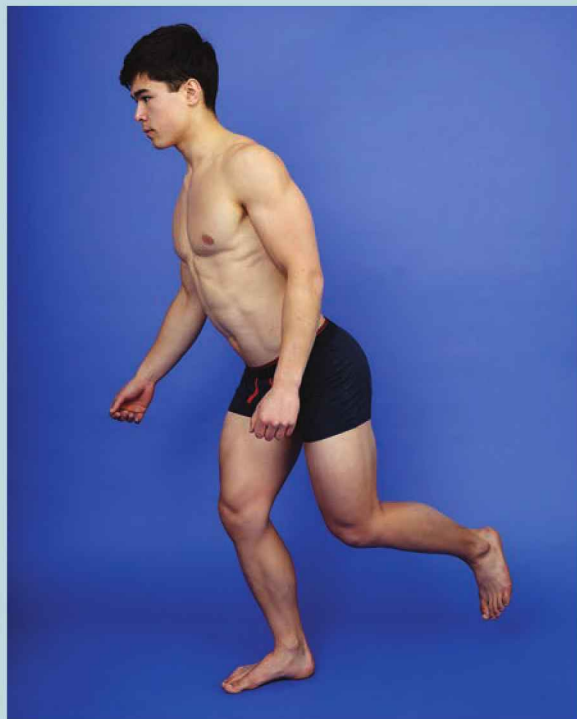
2. f. 'Fitter' training for 5 minutes.
2. g. Standing on the sliding board, side-to-side skating movements on the sliding board are done as five sets of 1-minute training periods with 15-second recovery.

Practical tips

- Supervision is important—the patient should be instructed by a physiotherapist, a clinician, an athletic trainer or another qualified person who has been trained in the details of the program.
- Exercises such as 1d and 2e are very important, especially at the end of the training period, but they are technically difficult.
- The athletes can do the program at home or at the gym or fitness club, but we recommend physiotherapist supervision for three to four times within the first 2 weeks

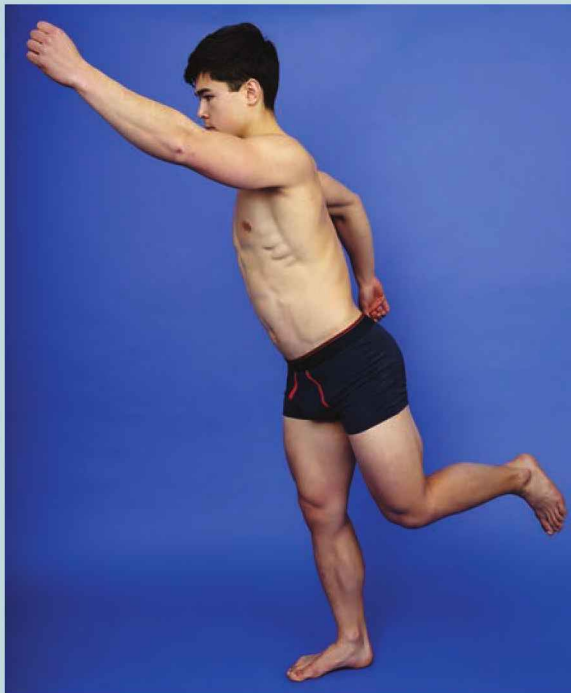


(a)



(b)

Figure 32.8 (a-c) Cross-country skiing. Dynamic hip flexion on one leg with coordinated arm swing



(c)

Figure 32.8 (cont.)

and after that a visit every 10–14 days to check the technique and ensure progression.

- Patience is the key to success.
- Patients often make good progress in the first few weeks, but symptoms can plateau from that period until the 6–9 week period, when there is a positive 'breakthrough'.
- It is important to use pain as a guide to how much to do. Muscle soreness similar to that after a regular practice in the sports field is not a problem, but if the patient experiences pain from the injury, the intensity of the exercises should be adjusted.
- Pain medication including NSAIDs should be avoided.
- Athletes should continue with some of the exercises on a regular basis (1–2 times a week) for at least a year after total recovery and return to sport.
- The athlete must appreciate that successful rehabilitation of longstanding groin pain takes a minimum of 8–12 weeks.

PRACTICE PEARL

An advance in recent years has been that rehabilitation programs have evolved from time-based protocols such as that used by Hölmich, to criteria-based protocols (see Chapter 18).

Reflecting this more contemporary approach, the following rehabilitation program is divided into four phases, each with a different rehabilitation focus. In general, the rehabilitation program works from focus and isolation on the specific injured tissues towards integration of the specific structures into sports-specific function and return to sport based on individual and continuous assessment of:

- injury (structural) status
- functional status
- performance status; and finally
- sports participation status, at specific time points during the rehabilitation period.

PRACTICE PEARL

The rehabilitation of acute groin injuries can be divided into four main phases prior to pre-injury competitive level:

1. acute/subacute phase
2. conditioning phase
3. sports-specific phase
4. return-to-sport phase.

1. Acute/subacute phase

The acute and subacute phase is characterised by musculotendinous pain and/or inflammation at the musculotendinous junction (MTJ), in the tendon or at the insertional enthesis. This phase focuses on the initiation of tissue repair and/or regeneration, but also includes effects of possible disuse as a result of injury. The primary goals to be addressed during this phase are protection of the injured structure, control of pain and inflammation, normalisation of flexibility within restrictions and preventing excessive muscular inhibition using isometric contractions with no or minimal pain provocation.

Criteria for progression to conditioning phase

The criteria for progression to the conditioning phase are:

- no or minimal pain with all exercises and clear muscle activation during isometric contractions
- active hip abduction range of movement (ROM) of at least 50% of the non-injured side.

2. Conditioning phase

This phase focuses on tissue repair and/or regeneration. In this phase gradual and progressive loading of adductor, hip flexor and abdominal muscles is commenced while still respecting the structural injury and underlying healing processes depending on the location of the injury (MTJ), tendon, aponeurosis or insertion) at this early stage.

Gentle dynamic muscle exercises are commenced starting off with a low load from 20–30 repetitions maximum (RM). Strength training machines, side-lying using the weight of the leg and standing exercises with elastic bands (Fig. 32.7) can be used for targeting the adductor longus muscle and at least two different exercise types should be used with 3×20 repetitions of each.

The musculotendinous conditioning should focus on concentric, isometric and eccentric contractions during these exercises of the specific injured tissues structure. For the abdominals, abdominal crunches in supine, frontal rolling exercises with Swiss ball, modified planks or training in a machine should also be performed using the same principles as applied for the adductors. For the hip flexors, isometric hip flexion in various degrees of range of motion, hip flexor training with elastic bands, and hip flexion work in a machine or with free weights should also be performed using the same principles as applied for the adductors and the abdominals. All these exercises should be performed 3 times a week with at least one day of rest in between sessions.

Pain during exercises must not be neglected, but some discomfort and pulling may be felt during early initiation of these exercises. A general rule of thumb is that pain and discomfort should be less than 3/ on a numerical rating scale (NRS) during exercise and that initial pain levels before exercise are reached shortly after exercises are finished. Increasing morning pain and stiffness, compared to the initial state, is also a good indicator that the exercises have been progressed too quickly. Once the patient is pain-free, pain-free walking can begin and be gradually increased in speed and distance.

The criteria for when the patient may return to running are when:

- brisk walking is pain-free
- resisted hip adduction testing and squeeze testing is pain-free
- there is minimal adductor guarding.

Criteria for progression to sports-specific phase

The criteria for progression to the sports-specific phase are:

- no or minimal pain with all phase 2 exercises and normal execution (no compensation strategies to avoid loading injured structures) of dynamic exercises of adductor, hip flexor and abdominal muscles
- active ROM at least 80% of the non-injured side and adductor strength of at least 60% of the non-injured side; which means that the athlete will be able to do
 - (i) 20 repetitions maximum using the same resistance based on 20 repetitions maximum from the uninjured side, and/or able to do 20 repetitions of exercises for adductors (side-lying adductor lifts), (ii) 20 repetitions for abdominals (abdominal crunches lying supine) and

(iii) 20 repetitions for hip flexors (hip flexions against your own hand lying supine) on the floor with proper form, and with no or minimal pain.

3. Sport-specific phase

The primary goals to be addressed during this phase are restoration of musculotendinous endurance and strength, aerobic and anaerobic capacity, and neuromuscular control/balance/coordination. Dynamic muscle strengthening exercises for the adductor and hip flexor muscles in standing, using elastic band resistance, are continued starting off with a load of 20 RM progressing towards 10 RM. The same principle is applied for abdominal exercises where a Swiss ball is used.

The musculotendinous conditioning should focus on slow heavy concentric, isometric and eccentric contractions during these exercises. Exercises on a sliding board (Fig. 32.6d) are commenced by introducing 5 to 10 sets of 1 minute of skating from side to side. All these exercises should be performed three times a week with at least one day of rest in between. Pain during exercises must not be neglected and the same guidelines accepting pain 0–2 on an NRS as stated earlier apply.

Various progressive running and kicking regimens can be used. An example is described below.

- The regime starts with 50–100 m run-throughs with 5–10 m acceleration and deceleration phases with walk recovery, depending on running demands of the specific sport. Patient should commence with 6–8 repetitions on alternate days. Key criteria (adductor guarding, squeeze test) should be assessed immediately after each session and again the next morning. The running program can be progressed further by replacing walk recovery with jog recovery. The aim should be to build up to 20 run-throughs and jog back.
- Lateral running (gradual change of direction such as figure of eight) can be commenced when the above running program is completed pain-free, the hip flexion test is still pain-free with no crossover sign, there is no adductor guarding and, in addition, the squeeze test is pain-free. Figure-of-eight running should commence slowly with very gradual change of direction, then gradually increase both speed and sharpness of changes of direction.
- Finally, Copenhagen hip adductor exercises (Fig. 32.9) should be incorporated starting at level 1 with 3–5 repetitions on each side (change top leg). Then progress to intermediate level 2 with 7–10 repetitions on each side and then level 3 with 12–15 repetitions on each side. All exercises above should be performed 2–3 times a week, with a day of rest from these exercises in between.

Initial agility drills should be performed with proper form, with no clear and visible compensation strategies



(a)



(b)

Figure 32.9 (a) Copenhagen adductor exercise
(b) A partner exercise where the player is lying on the side of the non-dominant leg with one forearm as support on the floor and the other arm placed along the body. The dominant leg is held at approximately the height of the hip of the partner, who is holding the leg with one hand supporting the ankle and the other supporting the knee. (Fig. 32.9a) The player then raises the body from the floor and the non-dominant leg is adducted so that the feet touch each other and the body is in a straight line (Fig. 32.9b)

during maximal and forceful actions such as accelerating, decelerating, twisting turning and kicking.

In kicking sports, short stationary kicking can be commenced when adduction and squeeze tests are pain-free without crossover. The player may gradually increase the kicking distance and then start shorter kicking on the run. The last stage in the kicking program is long kicks at full pace and kicking around the body.

Criteria for progression to return-to-sport phase

The criteria for progression to the return-to-sport phase are:

- no or minimal pain with all exercises and normal execution (no compensation strategies to avoid loading

injured structures) of dynamic exercises of adductor, hip flexor and abdominal muscles

- active ROM of nearly 100% of the non-injured side and adductor strength of at least 80% of the non-injured side, able to do 3×10 RM, by using the same resistance based on 10 RM from the uninjured side for hip adductors or flexors, or able to do 3×10 side-lying adductor lifts with external load of at least 5 kg, and able to do 3×10 hip flexor exercises lying supine using the weight of the leg as the external weight and 3×10 abdominal crunches lying over Swiss ball with proper form, and with no or minimal pain.

4. Return-to-sport phase

The return-to-sport phase is characterised by activities that focus on returning the athlete to full sporting function and participation, with successful return to previous functional and sporting level and the prevention of re-injury. The primary goals during this phase are maintenance of musculotendinous endurance and strength, aerobic and anaerobic capacity, and improving neuromuscular control/balance/coordination.

Dynamic muscle exercises using the elastic bands are continued with a load of 8–10 repetitions maximum, with focus on different and increasing speeds of eccentric contractions. Copenhagen adductors are continued at level 3 and exercises on a sliding board are continued by introducing 10–20 times 1 minute of skating from side to side. All these exercises should be performed 2–3 times a week with at least one day of rest from strength training in between.

Criteria for return to pre-injury competitive level

Participation in the specific sport, on an individual level or with the rest of the team, progressing from 30 minutes to 90 minutes of training and participating in 1–3 weeks of full training before full return to sport (match/competition activity) can be commenced, depending on the severity of the initial injury (the more severe the longer participation period before full return).

Key clinical signs suggestive of 'excessive loading' during rehabilitation

As the therapist must continually guard against the player 'overdoing' rehabilitation, we share the following signs that appear to suggest excessive loading and deterioration during rehabilitation:

- pain on passive hip abduction
- adductor muscle 'guarding' with increased muscle tone on passive combined hip external rotation and abduction
- pain and weakness with resisted adduction and squeeze testing.

Other non-surgical treatments for adductor-related groin pain

Compression shorts

Compression shorts can be used in cases where athletes continue to train and play despite ongoing symptoms. This can be either in cases where therapy does not result in a complete resolution of symptoms or where, for whatever reasons, athletes return to the field without having completed rehabilitation. They can also be useful when athletes are not fully confident after longstanding injuries.

Compression shorts reduce pain during athletic activities without a significant effect on performance. A study of 29 healthy athletes wearing two types of shorts used surface electromyography (EMG) of the adductor muscles during a 45° cutting movement. When wearing compression shorts with directional reinforcement properties, athletes had reduced adductor muscle activity during cutting.⁵² The reduced muscle activity may be one mechanism by which compression shorts reduce pain. The reductions in pain are only small and to what extent this may postpone recovery has not been studied.

Manual therapy

A case series followed by an RCT have studied the effects of a multimodal treatment program, including an adductor muscle manipulation, stretching and a graded running program, in longstanding adductor-related groin pain.^{53, 54} The RCT was evaluated as being high quality in a recent systematic review which concluded that multimodal treatment including manual adductor manipulation can result in a faster return to play, but not a higher treatment success, than a partially supervised active physical training program.¹³

There are a number of clinical scenarios in which manual therapy including adductor manipulation (Fig. 32.10a and b) or soft tissue therapy (Fig. 32.10c) can be considered. It can be offered as a standalone treatment, but should be considered more as an adjunct to active exercises in cases where:

- exercise therapy has not given a satisfactory result.
 - 'weekend warriors' are not motivated or able to perform exercise therapy for the required intensity and duration
- manual therapy might be of help.

Iliopsoas-related groin pain

Iliopsoas-related groin pain is the second most common injury in the groin region.³⁻⁵ It is characterised by pain in the anterior part of the proximal thigh, more laterally than adductor-related groin pain. Iliopsoas-related groin pain can present in isolation, but may also be part of the disturbed muscle balance in the region when other structures are injured. It is an important differential diagnosis for

hip-related groin pain; not only because of the pain location, but also since the iliopsoas with its close relation to the hip joint as an anterior stabiliser tends to get involved whenever there is a hip or groin problem. It seems to be a muscle that frequently gets involved in compensating for the overload.

Diagnostic criteria

The diagnostic criteria for iliopsoas-related groin pain are *iliopsoas tenderness* (Fig. 32.3i and j) and *iliopsoas-related groin pain is more likely if there is pain on resisted hip flexion AND/OR pain on stretching the hip flexors* (Fig. 32.3f).²²

Treatment

Most of the literature regarding the iliopsoas is related to the snapping hip and in recent years to hip arthroscopy. The painful iliopsoas muscle and/or tendon in athletes have not been the focus of any evidence-based research. The authors' preferred treatment of iliopsoas-related groin pain aims to strengthen the hip flexor muscles (Fig. 32.11) using isometric, concentric and eccentric contractions,⁵⁵ and combining this with a series of pelvic stabilisation and balance exercises.



(a)

Figure 32.10 Manipulation of the adductors (a) Using the left hand to hold the heel, the right hand takes a hold of the muscle belly of the adductors



(b)



(c)

Figure 32.10 (cont.) (b) The left hand brings the hip into forced abduction and external rotation to apply the maximal tolerable stretch to the adductor group, while the right hand controls the tension; the motion is circular and flowing
(c) Soft tissue therapy to the adductors



(a)



(b)

Figure 32.11 Hip strengthening exercises. Standing hip flexor strengthening with elastic band from start position (a) to finish position (b). Dynamic strength and coordination exercise with emphasis on maximal concentric, isometric and eccentric strength of hip flexor muscles. With elastic band fixated from above (here) the emphasis will be on rectus femoris and adductor longus in relevant kicking position, whereas if elastic band is fixated from below then the emphasis will be on strengthening the iliopsoas by focusing on optimal resistance towards full hip flexion

Additional physiotherapy, including stretching, soft tissue therapy and trigger point stimulation for symptomatic relief, might also be helpful. In most cases, players can return to play in 4–6 weeks. If the strength training is not progressing satisfactorily because it is too painful to perform the exercises, it is often possible to decrease the pain with an ultrasound guided local corticosteroid injection along the iliopsoas tendon.

Causes for groin pain can be seen in isolation or alongside each other in the same patient and the treatment needs to be directed at all entities that are diagnosed. The general exercises aiming at the coordination and synergies of the muscles and joints related to the hip and pelvis are along the

same lines in most programs. However, it is very important that the individual entities are addressed individually with specific strengthening and endurance exercises.

In the internal coxa saltans (snapping psoas) the patient feels a snap happening during activity in the groin region when the hip is extended from the flexed position. The snap can sometimes be audible. If the snapping is very uncomfortable or painful it can, in our experience, be treated with a strengthening and stretching program in the majority of cases. On rare occasions, a partial release of the tendon at or close to the insertion may be performed with good results. The surgery can be performed as an open or arthroscopic/endoscopic procedure.⁵⁶

Inguinal-related groin pain

Inguinal-related groin pain is probably one of the areas surrounded by the most confusion when dealing with groin pain in athletes. This entity has been defined in multiple ways, been given numerous differing nomenclatures and been the subject of much debate. Differing terminology has emerged from different geographical regions and the terms 'sports hernia', 'sportsman's hernia', 'sportsman's groin', 'posterior wall weakness', 'incipient hernia', 'Gilmore's groin' and 'hockey groin' have all been used in the past.²² A consensus statement from the British Hernia Society proposed the term 'inguinal disruption', which can be used when three out of five possible findings are present.⁵⁷ This statement was the subject of some debate as it introduced yet another term into the fray and was not part of a wider classification system.⁵⁸

Diagnostic criteria

The diagnostic criteria for inguinal-related groin pain are:

- *pain in inguinal canal region and tenderness of the inguinal canal* (Fig. 32.3k-l); *no palpable inguinal hernia is present*
- *more likely if aggravated with abdominal resistance* (Fig. 32.3g) or *Valsalva/cough/sneeze*.²²

There is no attempt made to describe any possible underlying pathology, as the exact substrate is still unknown. Bulging of the posterior wall, bulging causing entrapment neuropathy, tendinopathy of the inguinal ligament, tears of the external oblique aponeurosis and tearing of the conjoint tendon have all been proposed as being 'the pathology' underlying the problem. The authors feel that at this point in time, the issue of underlying pathology has yet to be resolved.

The confusion surrounding inguinal-related groin pain probably explains some of the difficulties when diagnosing and treating this condition. The actual prevalence remains unknown and this is clearly illustrated by the variety in reporting this in the literature.

A 1995 Australian case series from a sports physician on 189 cases of groin pain in athletes defined 'incipient inguinal hernia' as chronic pain above the pubic tubercle, increased by activities such as running or kicking,¹⁷ local tenderness over the conjoint tendon area and posterior inguinal wall laxity at surgical exploration. In this series, 50% of the athletes were found to have incipient inguinal hernia. This is in stark contrast to a series of 207 athletes from Hölmich in 2007, who defined 'sports hernia' to be the presence of tenderness of the external inguinal ring and conjoint tendon and close to its insertion on the pubic tubercle in the absence of a visible or palpable inguinal hernia.⁵⁹ In this series, 4 of the 207 athletes (2%) were classified as having 'sports hernia'.

A prospective study examining new cases of groin pain in soccer players over the course of a season in Denmark defined 'abdominal-related groin pain' as the presence of pain on palpation of the abdominal muscle insertion together with pain on abdominal flexion against resistance. Eleven of the 58 entities (19%) were abdominal-related and in 9 of the 11 cases this coexisted with the presence of adductor-related groin pain.¹⁵

Treatment

When considering treatment options for inguinal-related groin pain, it is important to consider the available evidence. The systematic review discussed earlier in the section 'Evidence for treatment' noted that there 'were no studies focused on the conservative treatment of sport's hernia with a well-described treatment protocol'.¹³ It should be noted that only case series with more than 10 subjects were included so some smaller previous series which report success were not included.^{60, 61}

In practice, we recommend to commence an active exercise-based approach with similar principles as those in the program earlier described for longstanding adductor-related groin pain. This approach aims to strengthen the abdominal muscles, including isometric, concentric and eccentric contractions, in combination with a series of adductor and abductor-strengthening exercises, pelvic stabilisation and balance exercises. The abdominal exercises used in the adductor program (Fig. 32.6b and c) can be used as a starting point,⁵⁰ progressing into more strenuous exercises for the abdominal wall through plank exercises (Fig. 32.12a) and then into modified plank exercises with high muscle activation of both internal and external obliques.⁶² A Swiss ball can also be used in the late stage conditioning phase before return to sport to mimic some of the external loads and forceful eccentric/concentric contractions that the abdominals go through, especially in kicking, throwing or shooting (ice hockey) (Fig. 32.12b and c).⁶³

As with the adductor program, the total length of the exercise training period is dependent on criteria-based progression, but is generally between 8–12 weeks. Absence from training is again necessary as the functionality and strength of the abdominal muscle group has to be re-established, and pain brought under control without continued provocation by the sport that first precipitated the injury. Stationary cycling and general fitness training not involving the abdominals can be allowed if they can be done without pain. Jogging is again allowed after 6 weeks as long as it does not provoke any groin pain.

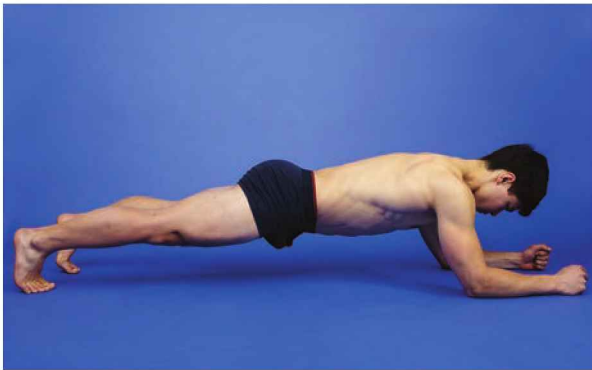
When the patient is able to jog and undergo his or her regular treatment without pain, the treatment program is completed. The athlete is allowed to gradually progress to demanding sports-specific training followed by sports

participation. No excessive stretching of the abdominal or the adductor muscles during this 2-3 month exercise training period is recommended as it can put unnecessary strain on the adductor and abdominal complex.

Surgery

In cases where conservative treatment is not successful or surgery is considered for other reasons, it should be noted that there have been no comparative studies on the superiority of different techniques. A study comparing laparoscopic to open hernia repair is currently ongoing.⁶⁴

A review identified 12 surgical studies on open hernia repair, 10 studies on laparoscopic hernia repair and 16 studies in which combined procedures (including adductor tenotomy) were performed.¹³ Additional neurectomy was performed in 12 studies. There was only one high-quality study identified in the review.⁶⁵ The

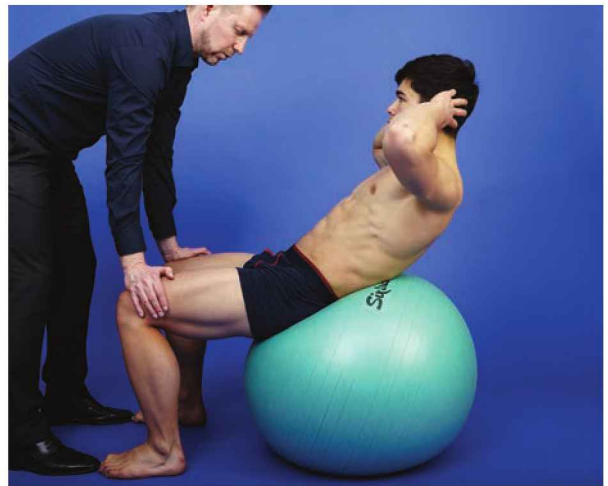


(a)



(b)

Figure 32.12 (a) Long lever planks with static maximal trunk muscle activation (b) Swiss ball curls



(c)

Figure 32.12 (cont.) (c) Dynamic abdominal strengthening with eccentric emphasis.

studies with the weakest methodological score reported the highest treatment success. This is important to take into consideration when using these published results to help inform patient decisions in clinical practice.

The one high-quality RCT compared laparoscopic hernia repair to conservative treatment in cases of 'sportsman's hernia (athletic pubalgia)'.⁶⁵ This was defined as athletes having more than 3 months history of pain above the inguinal ligament. There was pain on palpation of the deep inguinal ring and in cases where there was also pain on the pubic bone, pubic tubercle or adductors, or bone marrow oedema seen on the MRI, the athletes were also included.

The study randomised 60 athletes to treatment with either a bilateral totally extraperitoneal (TEP) laparoscopic hernia repair or conservative treatment. Unfortunately, the conservative treatment protocol is only described briefly as being total rest from their sports, active physiotherapy, steroid injections into the painful area and oral anti-inflammatory analgesics. This treatment was given 3 times a week for 90 minutes for 8 weeks and was supervised by the club physiotherapist.

This study found that 97% of those treated with surgery had returned to sport at 1 year compared to 50% of the conservative group. Until now this is the only high-quality study comparing surgery with conservative treatment. The poor description of the conservative treatment makes interpretation for clinical practice challenging, but it seems that some patients may benefit from conservative treatment and will be able to return to sport without having surgery. As complications from surgery are not uncommon in this region, non-surgical treatment should be considered and exhausted first.

The consensus agreement from the British Hernia Society noted that there was no proven best surgical technique.⁵⁷ This means that local availability along with the experience and preference of the surgeon are likely to be important when decisions are made as to which type of surgery to perform.

Inguinal hernia

Inguinal hernias occur in athletes as in the general population. They can be either direct or indirect. Small hernias may become painful as a result of exertion. Symptoms may include a characteristic dragging sensation to one side of the lower abdomen aggravated by increased intra-abdominal pressure, such as coughing. On examination, there is occasionally an obvious swelling and the hernia can be palpated. In cases of a symptomatic inguinal hernia, the treatment consists of surgical correction of the defect.⁶⁶

Pubic-related groin pain

No epidemiological data are available in the literature regarding how common pubic-related groin pain is.

Diagnostic criteria

The diagnostic criteria for pubic-related groin pain are:

- *local tenderness of the pubic symphysis and the immediately adjacent bone* (Fig. 32.3m)
- *no particular resistance tests to test specifically for pubic-related groin pain could be identified.*²²

Verrall et al. described clinical signs including pubic symphysis and superior pubic ramus tenderness as part of a bone stress-related problem.⁶⁷ The presence of BMO on MRI in the pubic bone was also a part of the diagnosis. The clinical criteria included the presence of at least 6 weeks of groin pain, located in the adductor and/or pubic bone region and present during and/or after sporting activity. No particular resistance tests to test specifically for pubic-related groin pain have been identified, but squeeze testing seems to be the most relevant.⁶⁸

In a study of 10 Australian rules football players with longstanding groin pain, using the criteria mentioned, biopsy was taken from the area in the superior pubic rami where MRI showed BMO. Histologically new woven bone was found and this was considered to represent a repair process that was the pathological stress reaction that was causing the groin pain. No controls with BMO but without symptoms were included and as such it is not known if the woven bone is a normal reaction to the stress the bone is receiving playing football or it is a sign of pathology.³⁹

Treatment

The treatment for this condition as described by Verrall et al. consists of mainly unloading the lower extremities and gradually increasing load again over a period of 3–4 months moving from rest from all weight-bearing running activities for 12 weeks over a period with stationary cycling, a stepping device and a gradual return to running activity.⁶⁹ The athletes were allowed to return to football training when 30 minutes of pain-free interval running was tolerated. There is no evidence-based treatment available.

COMPLETE ADDUCTOR AVULSIONS

Complete proximal adductor avulsions (Fig. 32.13) are not common, but in American football (National Football League) the mean time for return to play in players treated conservatively for an acute adductor longus rupture was reported to be 6 $\pm$ 3 weeks (range, 3–12 weeks).⁷⁰ A recent study of two different adductor ruptures in a soccer player suggests that adductor longus ruptures can have very different recovery times. In this study, normal hip adductor strength was obtained after 52 weeks for the first injury, but after 10 weeks for the second injury.⁷¹ This highlights the importance of measuring hip adductor strength by using a hand-held dynamometer, as this gives important information on the actual recovery process of the adductors, which cannot be obtained from diagnostic imaging.

LESS COMMON INJURIES

Obturator neuropathy

Obturator neuropathy is a fascial entrapment of the obturator nerve as it enters the adductor compartment

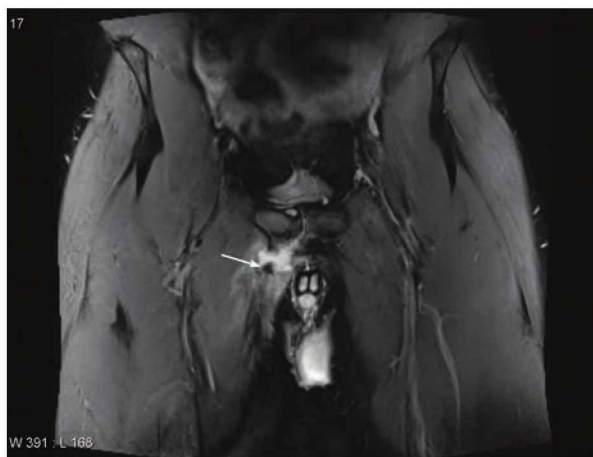


Figure 32.13 MRI coronal short tau inversion recovery (STIR) view of adductor avulsion. Images display acute rupture and retraction of adductor longus tendon on right side (arrow), with acute oedema (high intensity) and haemorrhage in surrounding muscle tissues

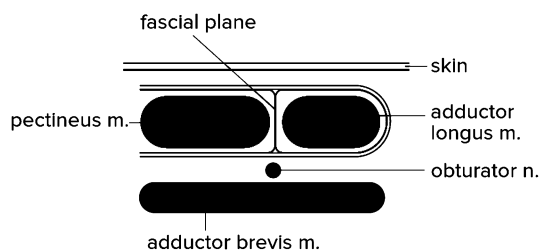


Figure 32.14 Obturator neuropathy—fascial arrangement

(Fig. 32.14). Obturator neuropathies have been reported to occur in Australian rules football and rugby. It has distinct clinical features that separate it from other causes of groin pain.^{72, 73}

Obturator neuropathy presents as exercise-related groin pain, which initially is located in the proximal groin but with increasing exercise radiates towards the distal medial thigh. There may be associated weakness or a feeling of a lack of propulsion of the limb during running but numbness is very rarely reported. At rest, examination findings can be non-specific, with pain on passive abduction of the hip, and pain and weakness on resisted hip adduction. The ipsilateral pubic tubercle is often tender. The essential component of the physical examination is to exercise the patient to a level that reproduces the symptoms followed by immediate examination of the patient. This examination will reveal weakness of resisted adduction and numbness over the distal medial thigh. The diagnosis is confirmed by needle EMG, which shows chronic denervation patterns of the adductor muscle group.

Conservative treatment of this condition, including sustained myofascial tension over the adductor compartment, neural stretches, spinal mobilisation and ilio-psoas soft tissue techniques, is generally unsuccessful.

The definitive treatment of this condition is surgical. An oblique incision is made in the proximal groin. The plane between the adductor longus and pectineus is identified and dissected, revealing the obturator nerve under the fascia over the adductor brevis. This fascia is divided and the nerve is freed up to the level of the obturator foramen. The fascial anatomy here is important, with the fascia of the adductor longus curving around the muscle medially and passing back deep to the muscle to become the fascia over the adductor brevis, which is thought to be responsible for the fascial entrapment of the obturator nerve. Post-surgical management includes wound management, soft tissue techniques and a graduated return to full activity over a period of 4–6 weeks.

Other nerve entrapments

A number of superficial nerves in the groin may become entrapped and should be considered as possible causes of

groin pain. The ilioinguinal nerve supplies the skin around the genitalia and the inside of the thigh and may produce pain as a result of entrapment. The genitofemoral nerve innervates an area of skin just above the groin fold. The lateral cutaneous nerve of the thigh is the most common nerve affected. This nerve supplies the outside of the thigh. This condition is known as 'meralgia paraesthetica' and is described in Chapter 33.

Pudendal nerve entrapment requires the presence of the following criteria: pain in the territory of the pudendal nerve (from the anus to the penis or clitoris); pain that is predominantly experienced while sitting; pain that does not wake the patient at night; pain with no objective sensory impairment; and pain relieved by diagnostic pudendal nerve block.⁷⁴

Treatment of these conditions is usually not necessary as they often spontaneously resolve. Meralgia paraesthetica is sometimes treated with a corticosteroid injection at the site where the nerve exits the pelvis, 1 cm medial to the anterior superior iliac spine. Occasionally, the nerve needs to be explored surgically and the area of entrapment released.

Stress fractures of the neck of the femur

Stress fracture of the neck of the femur is another cause of groin pain. The usual history is one of gradual onset of groin pain, poorly localised and aggravated by activity. Examination may show some localised tenderness but often there is relatively little to find other than pain at the extremes of hip joint movement, especially internal rotation. Plain radiograph may demonstrate the fracture if it has been present for a number of weeks but this investigation should not be relied on to rule out the condition; isotopic bone scan and MRI (Fig. 32.15) are the most sensitive tests.

Stress fractures of the neck of the femur occur on either the superior or tension side of the bone or on the inferior or compression side (Fig. 32.16). Stress fractures of the superior aspect of the femoral neck should be regarded as a surgical emergency and treated with either urgent internal fixation or strict rest. The concern is that such stress fractures have a tendency to go on to full fracture, which compromises the blood supply to the femoral head.

Stress fractures of the inferior surface of the femoral neck are more benign and can be treated with an initial period of non weight-bearing rest followed by a period of weight-bearing without running. They require at least 6 weeks of rest and usually considerably longer. Following the period of rest, a further 6 weeks of progressive loading will take the patient back to full training. Biomechanical (Chapter 8), nutritional and endocrine risk factors (Chapter 4) should be assessed and treated as appropriate.

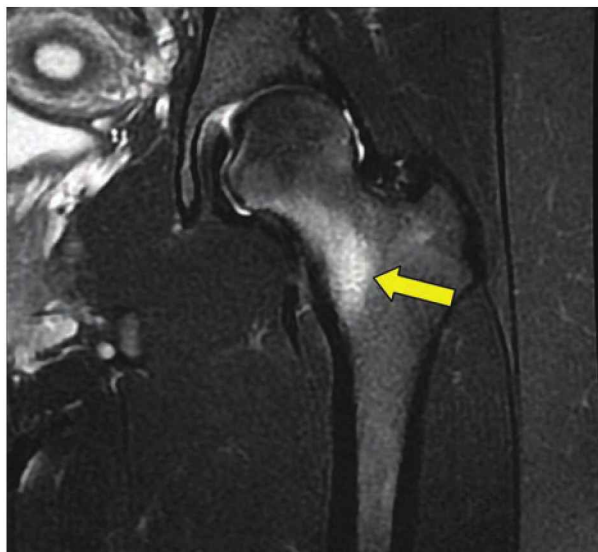


Figure 32.15 MRI showing stress fracture neck of femur
IMAGING IN SPORTS-SPECIFIC MUSCULOSKELETAL INJURIES, 2016,
P.741, ALI GUERMAZI, FRANK W ROEMER, MICHEL D CREMA. © SPRINGER
INTERNATIONAL PUBLISHING SWITZERLAND 2016. WITH PERMISSION OF
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Figure 32.17 MRI stress fracture of the inferior pubic ramus. Images show subcortical hypointense line (arrow) representing bony stress fracture parallel to the symphyseal joint surface in the inferior pubic ramus

Stress fracture of the inferior pubic ramus

Stress fracture of the inferior pubic ramus, especially in distance runners, is an important differential diagnosis of adductor tendinopathy. There is usually a history of overuse and localised tenderness, which is not aggravated by passive abduction or resisted adduction. In this condition, pain is often referred to the buttock.

A stress fracture may not be visible on plain radiograph for several weeks, whereas a radionuclide bone scan will

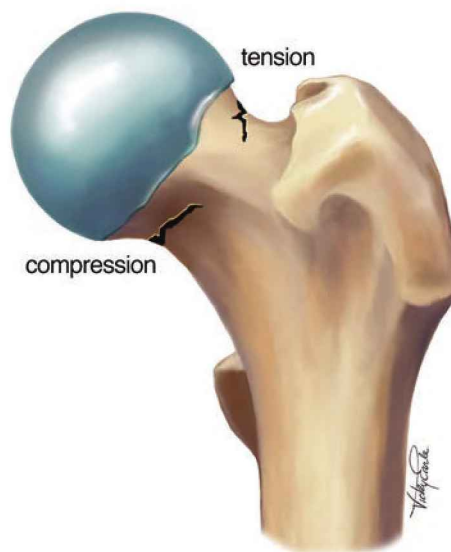


Figure 32.16 Stress fractures of the neck of femur—superior or tension fracture on the superior aspect of the femoral neck, and inferior or compression fracture on the inferior side

demonstrate a focal area of increased activity within hours and an MRI (Fig. 32.17) will show a focal area of bone oedema. As with any stress fracture, aetiological factors must be considered. Stress fractures of the inferior pubic ramus in females may be associated with reduced bone density, low initial aerobic fitness and nutritional insufficiency. Prolonged amenorrhoea is also linked with this stress fracture (Chapter 4).

Treatment consists of relative rest from aggravating activities such as running until there is no longer any local tenderness. Fitness should be maintained with swimming or cycling with gradual return to weight-bearing over a number of weeks. Predisposing factors such as a negative energy intake, muscular imbalance or biomechanical abnormality also require assessment and intervention.

Preventative strategies can be incorporated, especially in the female military/elite athlete population. Strategies may include pre-training interventions focusing on improving aerobic fitness to reduce fatigue fractures, and calcium and vitamin D supplementation.^{75, 76} Female recruits who report no menses during the year previous to recruitment should also be observed closely.⁷⁶

Referred pain to the groin

The possibility of referred pain to the groin should always be considered, especially when there is little to find on local examination. A common site of referral to the groin is the sacroiliac joint (SIJ) and this should always be assessed in any examination of a patient with groin pain. The SIJ may also refer pain to the scrotum in males

and labia in females. The assessment and treatment of sacroiliac problems are discussed in Chapter 30.

The lumbar spine may refer pain to the groin. The lumbar spine and thoracolumbar junction should always be examined in a patient with groin pain. Neurodynamic tests, such as the slump and neural Thomas tests, should be performed as part of the assessment (Chapter 15). Variations such as the addition of adduction or hip rotation may reproduce the patient's pain.

A positive neurodynamic test result requires further evaluation to determine the site of the abnormality. The position of reproduction of pain can be used to correct neural tightness by stretching. Active trigger points may also refer to the groin and should be treated with soft tissue therapy.

As mentioned in the section 'Terminology and definitions' there are many other conditions that can give rise to groin pain. When athletes present with groin pain that is not typical in nature, has no clear relationship with loading, nocturnal pain or pain associated with other non-musculoskeletal symptoms, one must think broadly in terms of possible differential diagnosis.

PREVENTION OF GROIN INJURIES

By late 2016, seven RCTs had investigated prevention of sports injuries where data on groin injuries had been included. These seven studies were evaluated and analysed in a systematic review and the data was pooled in a meta-analysis.⁷⁷

The studies examined the effect of different prevention interventions, predominantly looking at strengthening, coordination and balance exercises, and there were more than 2000 subjects in control and intervention groups, respectively. In total 157 injuries occurred, 68 in the intervention group and 89 in the controls. While the reduction is around 20%, this was not statistically significant. This means that at present, despite the number of trials examining groin injury prevention, there is no proven effective intervention. Due to the use of time loss definitions in the majority of studies, many overuse injuries will not have been registered, which is a limitation in the existing literature. Furthermore, compliance was not recorded in three studies and in the other four, compliance was less than intended.

Despite these limitations, including an active exercise strategy for strengthening and improving muscle function, load transfer and tolerance in the pelvic region may induce a relevant reduction of groin injuries. No reports of adverse events in relation to such a strategy have been documented. Clinicians and athletes should weigh up the time and effort that needs to be invested against the possible reduction to decide whether to implement in their setting.

Possible prevention strategies

A number of measures can be considered in clinical practice. Given the fact that previous injury is a risk factor, athletes with a history of groin pain should be given extra attention. Research on soccer players during the start of a new season has shown that those who had groin pain in the previous season still had reduced function and residual symptoms after the summer break.⁷⁸ In this light, it would seem prudent for athletes with groin pain in season to use the off-season to concentrate on treatment and recovery rather than simply resting.

A careful examination should be done at the end of the season, as described in the 'Clinical overview' section. Any deficits found can be used to target improvements in the off-season and should be combined with sports-specific training. A study from elite ice hockey showed that those players who performed the least sports-specific training in the off-season had triple the risk for a groin injury in the next season.⁷⁹

Little is known about the specific relationship between training load and groin pain. Recent literature from other fields has highlighted that athletes who have a rapid rise in training load, without sufficiently accumulating training to have ensured readiness are prone to injury. As such, careful consideration of training load can be part of groin injury prevention, but is likely to be part of a more general prevention strategy than only aimed at the hip or groin.

Monitoring players for changes in hip muscle strength in an attempt to predict the onset of groin pain was performed in a study on Australian rules football players.⁸⁰ Elite junior players ($n = 86$, 16–18 years old), were measured using squeeze tests on a weekly basis. They were also monitored for the presence of groin pain in the previous week. When they reported having groin pain 2 weeks in a row, this was considered to be the start of a groin injury.

The variation of the squeeze strength scores across the season was 2–3%. Twelve players developed groin pain and their strength in the week before the onset of the pain was reduced by $12 \pm 3\%$ from their own baseline. As such, monitoring of strength and the onset of groin pain can be used to allow early detection and hopefully aid prevention. Furthermore, in those footballers where groin pain is encountered through continuous screening and examination, early detection and management seem to be highly relevant as a groin pain duration of only 1–2 weeks is associated with a much better HAGOS score at the beginning of the pre-season/new season, compared to individuals with longer durations of groin pain in the previous season.⁷⁸

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e